Take-Home Project for Control Systems

Quadcopter Control: Tasks

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Instructions on Submission:

- Please submit your deliverables through Canvas in the "Assignment" tab.
- Deliverables:
  - A single report (a PDF file named `CS_TH_SURNAME.pdf`) containing your answers and plots. They are explicitly specified by **Deliverables** in this document.
  - An `.ipynb` file named `CS_TH_SURNAME.ipynb` containing your Python codes.

- Please mention your name and ID on the cover page of the report.

Notes:

- All the necessary details of quadcopters can be found in the document "`quadcopter_intro.pdf`".
- Please use the Jupyter Notebook "`quadcopter.ipynb`" to complete the following tasks.
- Please follow the instructions properly, otherwise points will be deducted.
- Students are expected to submit independent solutions.
- Plagiarism is prohibited and might lead to unpleasant consequences.

T1: Linearization [2P]

Suppose that the quadcopter is hovering at a fixed position $[x_d,\ y_d,\ z_d]^\top$ with a heading ψ_d radians (i.e. $[x,\ y,\ z,\ \psi]^\top = [x_d,\ y_d,\ z_d,\ \psi_d]^\top$). In this case, the translational velocity is $[\dot{x},\ \dot{y},\ \dot{z}]^\top = [0,\ 0,\ 0]^\top$ and the rotational velocity is $[p,\ q,\ r]^\top = [0,\ 0,\ 0]^\top$. In addition, the roll and pitch angles are equal to zero (i.e., $[\phi,\ \theta]^\top = [0,\ 0]^\top$). The thrust force u_1 is equal to the quadcopter's weight force mg so that the quadcopter does not fall. The rolling, pitching, and yawing moments are equal to zero (i.e. $u_2 = u_3 = u_4 = 0$) so that the quadcopter neither rotates nor moves. In other words, the equilibrium point of the quadcopter is given by

$$\begin{aligned} [x,\ y,\ z,\ \dot{x},\ \dot{y},\ \dot{z},\ \phi,\ \theta,\ \psi,\ p,\ q,\ r]^\top &\overset{\text{equilibrium}}{=} [x_d,\ y_d,\ z_d,\ 0,\ 0,\ 0,\ 0,\ 0,\ \psi_d,\ 0,\ 0,\ 0]^\top, \\ [u_1,\ u_2,\ u_3,\ u_4]^\top &\overset{\text{equilibrium}}{=} [mg,\ 0,\ 0,\ 0]^\top. \end{aligned} \tag{5}$$

i) Complete the corresponding code fragment in the Jupyter Notebook "`quadcopter1.ipynb`" to linearize the system (1), (3), and (4) in the document "`quadcopter_intro.pdf`" around the equilibrium point (5).

Consider the parameters with the values listed in the following table.

Mass (m)	$0.03\ [kg]$	Drag Force Coefficient (k_i) $i \in \{x, y, z\}$	$4.5 \times 10^{-3}\ [kg/s]$
Gravitational Acceleration (g)	$9.81\ [m/s]$	Drag Moment Coefficient (k_i) $i \in \{p, q, r\}$	$4.5 \times 10^{-4}\ [kg \cdot m^2/s^2]$
Inertia (I_x, I_y)	$1.5 \times 10^{-5}\ [kg \cdot m^2]$	Lateral Equilibrium (x_d)	$1\ [m]$
Inertia (I_z)	$3 \times 10^{-5}\ [kg \cdot m^2]$	Longitudinal Equilibrium (y_d)	$1\ [m]$
Heading Equilibrium (ψ_d)	$\pi/4\ [rad]$	Altitude Equilibrium (z_d)	$1\ [m]$

Note: Consider the deviation state vector $[\Delta x,\ \Delta y,\ \Delta z,\ \Delta\dot{x},\ \Delta\dot{y},\ \Delta\dot{z},\ \Delta\phi,\ \Delta\theta,\ \Delta\psi,\ \Delta p,\ \Delta q,\ \Delta r]^\top = [x,\ y,\ z,\ \dot{x},\ \dot{y},\ \dot{z},\ \phi,\ \theta,\ \psi,\ p,\ q,\ r]^\top - [x_d,\ y_d,\ z_d,\ 0,\ 0,\ 0,\ 0,\ 0,\ \psi_d,\ 0,\ 0,\ 0]^\top$, the deviation input vector $[\Delta u_1,\ \Delta u_2,\ \Delta u_3,\ \Delta u_4]^\top = [u_1,\ u_2,\ u_3,\ u_4]^\top - [mg,\ 0,\ 0,\ 0]^\top$, and the deviation output vector $[\Delta x,\ \Delta y,\ \Delta z,\ \Delta\phi]^\top$. Make sure that the resulting linear differential equations correspond to the state space model whose matrices are given by

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{k_x}{m} & 0 & 0 & g\sin(\psi_d) & g\cos(\psi_d) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{k_y}{m} & 0 & -g\cos(\psi_d) & g\sin(\psi_d) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{k_z}{m} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{k_p}{I_x} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{k_q}{I_y} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{k_r}{I_z} \end{bmatrix},$$

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{m} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \frac{1}{I_x} & 0 & 0 \\ 0 & 0 & \frac{1}{I_y} & 0 \\ 0 & 0 & 0 & \frac{1}{I_z} \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Please print the results of the system matrices A, B in your submitted Jupyter Notebook `CS_LE1_SURNAME.ipynb`. This part does not need to be included in the report.

Hints: Use `symbols,jacobian,subs` in the `sympy` package.

T2: Stability Analysis [2P]

Consider the system described by the aforementioned linearized state-space model.

i) Complete the corresponding code fragment in the Jupyter Notebook "`quadcopter1.ipynb`" to compute the eigenvalues of the linearized system matrix A, and then find its Jordan form J.

   Hints: Use `linalg.eigvals` in the `numpy` package and `jordan_form` in the `sympy` package.

ii) Is the system stable or unstable? If stable, is it asymptotically stable?

Deliverables:

i) The eigenvalues and the Jordan form of the linearized system matrix A.

ii) An answer (with justification) to the question if the model is unstable, stable or asymptotically stable (in a couple of sentences).

T3: Forced Responses [1.5P]

Consider the linearized rolling dynamics only (i.e. the state space model whose states are $\Delta\phi$ and Δp, input is Δu_2, and output is $\Delta\phi$).

i) Complete the corresponding code fragment in the Jupyter Notebook "`quadcopter1.ipynb`" to simulate and plot the quadcopter's roll angle trajectory (i.e., $\Delta\phi$ vs. time) with zero initial condition (i.e., $\Delta\phi = 0$ and $\Delta p = 0$) for 10 seconds in the following cases.

   1) Impulsive input: $\Delta u_2(t) = 1 \times 10^{-5}\delta(t)$.
   2) Step input: $\Delta u_2(t) = 1 \times 10^{-5}\sigma(t)$.
   3) Sinusoidal input: $\Delta u_2(t) = 1 \times 10^{-5}\sin(t)$.

   Hints: Use `impulse_response`,`step_response`,`forced_response` in the `control` package.

Deliverables:

i) Three plots of forced responses.

T4: State-feedback Control [2P]

We consider the case where the heading angle $\psi_d = 0$, and we assume that the input u_3 (i.e., the pitching moment) is accessible. In this case, the x-axis dynamics are fully decoupled, and it can be described by the SISO sub-system with state vector $[x,\ \dot{x},\ \theta,\ q]^\top$ (note that to get a close connection with the experimental system, we use the notation of original variables instead of their deviations $[\Delta x,\ \Delta\dot{x},\ \Delta\theta,\ \Delta q]^\top$ in this learning element, which is justified since we consider the equilibrium point to be at the origin $[0,\ 0,\ 0,\ 0]^\top$, hereinafter the same), input u_3, output x, and state-space matrices

$$A_x = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{k_x}{m} & g & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{k_q}{I_y} \end{bmatrix}, \quad B_x = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{I_y} \end{bmatrix}, \quad C_x = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}, \quad D_x = 0, \tag{1}$$

where $m = 0.03$ kg, $k_x = 0.015$ kg/s, $k_q = 4.5 \times 10^{-4}$ kg$\cdot$ m^2/s^2, $I_y = 1.5 \times 10^{-5}$ kg$\cdot$m^2, and $g = 9.81$ m/s. *Note that the value of k_x is different than the one used in T1.* We consider the following problems.

i) Is the above system stable? Is it controllable?

ii) We would like to ensure that the closed-loop poles are -100, -50, -10, and -10. Please design a state feedback controller $u_3 = -k_1 x - k_2 \dot{x} - k_3 \theta - k_4 q$ to achieve this goal, where k_1, k_2, k_3, k_4 are parameters. Please provide the detailed steps.

Hints: Use `matmul`, `polymul`, `block`, `linalg.eigvals`, and `linalg.matrix_rank` in the `NumPy` package, `jordan_form` in the `SymPy` package, and `ctrb` and `canonical_form` in the `control` package. Note that the controllable canonical form returned by `canonical_form` is as follows, which slightly differs from the lecture.

$$\dot{\xi} = \underbrace{\begin{bmatrix} -a_{n-1} & -a_{n-2} & -a_{n-3} & \dots & -a_0 \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \dots & \vdots \\ 0 & \dots & \dots & 1 & 0 \end{bmatrix}}_{\triangleq A_\xi} \xi + \underbrace{\begin{bmatrix} 1 \\ 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix}}_{\triangleq B_\xi} u_3,$$

Deliverables:

i) Conclusions and explanation of whether the system is stable and controllable.

ii) Steps of designing a state-feedback controller for pole placement, as well as the form of this controller.

T5: Augmented State Feedback Control [4P]

Again, we consider the case where the heading angle $\psi_d = 0$, and we assume that the input u_3 (i.e., the pitching moment) is accessible. Let the state be denoted by $[x,\ \dot{x},\ \theta,\ q]^\top$. The state-space matrices A_x, B_x, C_x, D_x of the corresponding SISO sub-system are given by

$$A_x = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{k_x}{m} & g & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{k_q}{I_y} \end{bmatrix}, \quad B_x = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{I_y} \end{bmatrix}, \quad C_x = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}, \quad D_x = 0, \tag{2}$$

where $m = 0.03$ kg, $k_x = 0.015$ kg/s, $k_q = 4.5 \times 10^{-4}$ kg$\cdot$ m^2/s^2, $I_y = 1.5 \times 10^{-5}$ kg$\cdot$m^2, and $g = 9.81$ m/s. *Note that the value of k_x is again different than the one used in T1.*

We consider the following design problems.

i) Please design an integral control in addition to the state feedback control from T4 to ensure that the output (i.e., $C_x[x,\ \dot{x},\ \theta,\ q]^\top = x$) tracks a reference step signal $r(t) = 1$. Please plot the reference r and the trajectory x vs time. The simulated time is 10 seconds.

ii) Please design a feedforward control in addition to the state feedback control from T4 to ensure that the output (i.e., $C_x[x,\ \dot{x},\ \theta,\ q]^\top = x$) tracks a reference step signal $r(t) = 1$. Please plot the reference r and the trajectory x vs time. The simulated time is 10 seconds.

Hints: Use `step_response` in the `control` package.

Deliverables:

i) Steps and the design of an integral controller for reference tracking. A figure that shows the reference r and the trajectory x vs time.

ii) Steps and the design of a controller combining feedback & feedforward for reference tracking. A figure that shows the reference r and the trajectory x vs time.

T6: Observer-based Control [2P]

We assume that $\psi_d = 0$. The quadcopter dynamics along the x-axis are decoupled and given by

$$\dot{v} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -k_x/m & g & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -k_q/I_y \end{bmatrix}}_{\triangleq A_x} v + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/I_y \end{bmatrix}}_{\triangleq B_x} u, \quad x = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}}_{\triangleq C_x} v \tag{3}$$

with the state vector $v = [x\ \dot{x}\ \theta\ q]^\top$, the input $u = u_3$, and the output x. The parameters are $m = 0.03$ kg, $k_x = 4.5 \times 10^{-3}$ kg/s, $k_q = 4.5 \times 10^{-4}$ kg·m^2/s^2, $g = 9.81$ m/s, and $I_y = 1.5 \times 10^{-5}$ kg·m^2. *Note that the value of k_x is again different than the one used in T1.* Please design a state observer-based state feedback controller to drive the system (3) to the origin. The control design should satisfy the following objectives:

i) the state feedback is with closed-loop poles at $\{-0.3, -0.4, -10, -20\}$;

ii) the observer is with estimation poles at $\{-23, -26, -27, -28\}$;

The initial conditions of the quadcopter and the observer are $v = [-2\ 0\ 0\ 0]^\top$ and $\hat{v} = [0\ 0\ 0\ 0]^\top$, respectively. The simulation time is 10 s.

Hints: Use `place` and `initial_response` in the `control` package.

Deliverables:

i) The expression of your controller. Please explicitly mention the values of the control gain matrix K and the observer gain matrix L.

ii) A figure with legends showing the actual output trajectory x and the estimation error $x - \hat{x}$.